Variation of Parameters for a Scalar Equation

$$y' + p(t)y = q(t) \qquad y_1' = -p(t)y_1 \qquad y_c = cy_1$$

Variation of parameters replaces the constant c by a function $u(t)$.

$$y = uy_1 \qquad y' + p(t)y = u'y_1 + u\underbrace{(y_1' + p(t)y_1)}_{0}$$

So $u' = q(t)/y_1(t)$. Integrate and use the initial condition.

Example $y' + 2y = e^{-2t}$, $y(0) = 1$

Here $y_1(t) = e^{-2t}$, so $u' = 1$ and $u(0) = 1$. Then $u(t) = 1 + t$ and $y(t) = (1 + t)e^{-2t}$.

Variation of Parameters for a System

For $\boldsymbol{y}' = \boldsymbol{A}\boldsymbol{y} + \boldsymbol{f}(t)$, the **fundamental matrix** has n linearly independent complementary solutions as its columns.

$$\Phi(t) = [\, \boldsymbol{y}_1(t) \ \cdots \ \boldsymbol{y}_n(t) \,] \qquad \Phi' = \boldsymbol{A}\Phi \qquad \boldsymbol{y}_c = \Phi \boldsymbol{c}$$

Replace $\boldsymbol{c}$ by $\boldsymbol{u}(t)$ and substitute $\boldsymbol{y} = \Phi \boldsymbol{u}$.

$$\underbrace{(\Phi' - \boldsymbol{A}\Phi)}_{0}\boldsymbol{u} + \Phi \boldsymbol{u}' = \boldsymbol{f}(t) \qquad \boldsymbol{u}' = \Phi^{-1}\boldsymbol{f}(t)$$

At each t, the last equation amounts to solving $\Phi(t)\boldsymbol{u}'(t) = \boldsymbol{f}(t)$.

Initial Conditions and the Forcing Integral

Given $\boldsymbol{y}(0) = \boldsymbol{y}_0$, the coefficient vector starts at $\boldsymbol{u}(0) = \boldsymbol{\Phi}(0)^{-1}\boldsymbol{y}_0$.

$$\boldsymbol{y}(t) = \underbrace{\boldsymbol{\Phi}(t)\boldsymbol{\Phi}(0)^{-1}\boldsymbol{y}_0}_{\text{response to initial data}} + \underbrace{\boldsymbol{\Phi}(t)\int_0^t \boldsymbol{\Phi}(s)^{-1}\boldsymbol{f}(s)\,ds}_{\text{response to forcing}}$$

The first term makes $\boldsymbol{y}(0) = \boldsymbol{y}_0$ since the integral is zero at $t = 0$. For $\boldsymbol{A}$ constant

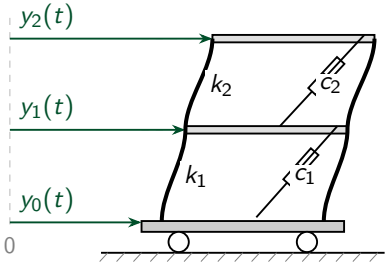
$$\boldsymbol{\Phi}(t) = e^{\boldsymbol{A}t} = \sum_{k=0}^{\infty} \frac{(\boldsymbol{A}t)^k}{k!} = \boldsymbol{I} + \boldsymbol{A}t + \frac{(\boldsymbol{A}t)^2}{2!} + \cdots$$

and the solution of the IVP is

$$\boldsymbol{y}(t) = e^{\boldsymbol{A}t}\boldsymbol{y}_0 + \int_0^t e^{\boldsymbol{A}(t-s)}\boldsymbol{f}(s)\,ds$$

Two-Story Building

Displacements: $\boldsymbol{y}(t) = (y_1(t), y_2(t))$ (m), t in seconds.



$m_2 = 3\,\mathrm{kg}$
$k_2 = 3\,\mathrm{N/m}$
$c_2 = 1\,\mathrm{N\,s/m}$

$m_1 = 4\,\mathrm{kg}$
$k_1 = 4\,\mathrm{N/m}$
$c_1 = 2\,\mathrm{N\,s/m}$

$y_0(t) = \sin(t)\ \mathrm{m}$

Worked problem

The base motion $y_0(t)$ is prescribed. The two floor displacements are measured to the right from rest.

$$\boldsymbol{y}(0) = (0, -1)\,\mathrm{m}$$
$$\boldsymbol{y}'(0) = (0, 2)\,\mathrm{m/s}$$

Story Forces and Floor Balances

Story i connects levels $i-1$ and i. The force on the upper level is $-f_i$ and the force on the lower level is $+f_i$, where

$$f_i = k_i(y_i - y_{i-1}) + c_i(y_i' - y_{i-1}')$$

$$\underbrace{\frac{\mathrm{N}}{\cancel{\mathrm{m}}}\cancel{\mathrm{m}} = \mathrm{N}}_{\text{spring}} \qquad \underbrace{\frac{\mathrm{N}\,\cancel{\mathrm{s}}}{\cancel{\mathrm{m}}}\frac{\cancel{\mathrm{m}}}{\cancel{\mathrm{s}}} = \mathrm{N}}_{\text{damper}} \qquad \mathrm{kg}\,\frac{\mathrm{m}}{\mathrm{s}^2} = \mathrm{N}$$

Substituting the story forces into Newton's law gives

$$\begin{aligned} 4y_1'' = -f_1 + f_2 \quad &= -4(y_1 - y_0) - 2(y_1' - y_0') + 3(y_2 - y_1) + (y_2' - y_1') \\ 3y_2'' = -f_2 \quad &= \qquad\qquad - 3(y_2 - y_1) - (y_2' - y_1') \end{aligned}$$

Floor Balances in Matrix Form

Collect the floor terms on the left and the prescribed base motion on the right.

$$\underbrace{\begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}}_{\boldsymbol{M}} \begin{pmatrix} y_1'' \\ y_2'' \end{pmatrix} + \underbrace{\begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix}}_{\boldsymbol{C}} \begin{pmatrix} y_1' \\ y_2' \end{pmatrix} + \underbrace{\begin{pmatrix} 7 & -3 \\ -3 & 3 \end{pmatrix}}_{\boldsymbol{K}} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 4y_0 + 2y_0' \\ 0 \end{pmatrix}}_{\boldsymbol{b}(t)}$$

$$\boldsymbol{M}\boldsymbol{y}'' + \boldsymbol{C}\boldsymbol{y}' + \boldsymbol{K}\boldsymbol{y} = \boldsymbol{b}(t) \qquad \boldsymbol{b}(t) = \begin{pmatrix} 4y_0(t) + 2y_0'(t) \\ 0 \end{pmatrix}$$

The units of $\boldsymbol{M}$, $\boldsymbol{C}$, $\boldsymbol{K}$, and $\boldsymbol{b}$ are kg, $\mathrm{N\,s/m}$, $\mathrm{N/m}$, and N, respectively.

For this example, the prescribed base motion is $y_0(t) = \sin(t)$.

First-Order Building IVP

Let $\boldsymbol{z} = (y_1, y_2, y_1', y_2')$. The top two equations give velocity. The bottom two give acceleration.

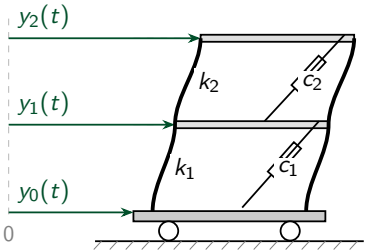
$$\boldsymbol{z}' = \boldsymbol{A}\boldsymbol{z} + \boldsymbol{g}(t) \qquad \boldsymbol{A} = \begin{pmatrix} \boldsymbol{0} & \boldsymbol{I} \\ -\boldsymbol{M}^{-1}\boldsymbol{K} & -\boldsymbol{M}^{-1}\boldsymbol{C} \end{pmatrix} \qquad \boldsymbol{g}(t) = \begin{pmatrix} \boldsymbol{0} \\ \boldsymbol{M}^{-1}\boldsymbol{b}(t) \end{pmatrix}$$

$$\boldsymbol{A} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -7/4 & 3/4 & -3/4 & 1/4 \\ 1 & -1 & 1/3 & -1/3 \end{pmatrix} \quad \boldsymbol{g}(t) = \begin{pmatrix} 0 \\ 0 \\ (4y_0(t) + 2y_0'(t))/4 \\ 0 \end{pmatrix} \quad \boldsymbol{z}(0) = \begin{pmatrix} 0 \\ -1 \\ 0 \\ 2 \end{pmatrix}$$

Dividing each force by its floor mass gives acceleration in $\mathrm{N/kg} = (\cancel{\mathrm{kg}}\,\mathrm{m/s^2})/\cancel{\mathrm{kg}} = \mathrm{m/s^2}$.

Two-Story Building Practice $My'' + Cy' + Ky = b(t)$

Displacements: $\boldsymbol{y}(t) = (y_1(t), y_2(t))$ (m), t in seconds.



$m_2 = 2\,\mathrm{kg}$
$k_2 = 4\,\mathrm{N/m}$
$c_2 = 2\,\mathrm{N\,s/m}$

$m_1 = 3\,\mathrm{kg}$
$k_1 = 6\,\mathrm{N/m}$
$c_1 = 3\,\mathrm{N\,s/m}$

$y_0(t) = 2\sin(t)\ \mathrm{m}$

$$\boldsymbol{y}(0) = (0, 1)\,\mathrm{m}$$
$$\boldsymbol{y}'(0) = (-1, 0)\,\mathrm{m/s}$$

1. Find $\boldsymbol{M}$, $\boldsymbol{C}$, $\boldsymbol{K}$, and $\boldsymbol{b}(t)$.
2. Write $\boldsymbol{z}' = \boldsymbol{A}\boldsymbol{z} + \boldsymbol{g}(t)$ and $\boldsymbol{z}(0)$.

Stop once the IVP is set up.

Practice Check $My'' + Cy' + Ky = b(t)$

$$\boldsymbol{M} = \begin{pmatrix} 3 & 0 \\ 0 & 2 \end{pmatrix} \quad \boldsymbol{C} = \begin{pmatrix} 5 & -2 \\ -2 & 2 \end{pmatrix} \quad \boldsymbol{K} = \begin{pmatrix} 10 & -4 \\ -4 & 4 \end{pmatrix} \quad \boldsymbol{b}(t) = \begin{pmatrix} 12\sin(t) + 6\cos(t) \\ 0 \end{pmatrix}$$

$$\boldsymbol{A} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -10/3 & 4/3 & -5/3 & 2/3 \\ 2 & -2 & 1 & -1 \end{pmatrix} \quad \boldsymbol{g}(t) = \begin{pmatrix} 0 \\ 0 \\ 4\sin(t) + 2\cos(t) \\ 0 \end{pmatrix} \quad \boldsymbol{z}(0) = \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}$$

As a check, the base term in the acceleration equation is $\big(6y_0(t) + 3y_0'(t)\big)/3$.